

Advancements in Perovskite Solar Cells: Power Conversion Efficiency and Operational Stability

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ABSTRACT

Perovskite solar cells have emerged as leading photovoltaic technologies due to rapid improvements in power conversion efficiency. We investigate compositional engineering of organic-inorganic halide perovskites and novel electron transport layers. Under continuous solar irradiation and thermal stress, passivated tandem cells demonstrate high stability and 28.5 percent certified conversion efficiency.

Keywords: Renewable Energy, Perovskite Solar Cells, Photovoltaics, Power Conversion Efficiency, Clean Energy

1. Introduction

Transitioning to zero-carbon energy systems requires highly efficient, low-cost solar photovoltaics. Metal halide perovskites combine strong optical absorption, long carrier diffusion lengths, and low-temperature solution processability.

2. Experimental Synthesis

We synthesized mixed-cation perovskite thin films incorporating formamidinium and cesium halides. Surface 2D/3D heterojunction passivation was applied to suppress non-radiative recombination at defect sites.

3. Stability & Efficiency Results

The fabricated cells achieved a power conversion efficiency of 28.5% with negligible hysteresis. Continuous operational testing under 1-sun illumination retained 92% of initial efficiency over 2,000 hours.

4. Conclusion

Perovskite photovoltaics offer a viable pathway toward scalable, cost-competitive clean electricity generation to mitigate greenhouse gas emissions.